

ELEC-1107 – Project Expectation, Clarification and Rubric

(Project: Powered Speaker and Mobile Device Charging Station)

The project will be a phased approach to applying into practice many of the concepts studied in this electronic devices course. In order to study some “real life” circuit applications, the students will perform design validation and component selection calculations for the project designs. The project involves building a 5 VDC, 3 Ampere USB (Universal Serial Bus) power supply for charging mobile devices. It will also incorporate a powered audio amplifier with a 1/8” audio jack input with a built in speaker. Finally, it will also include some basic safety and monitoring circuits.

The project will be split into 3 sections with a total weight of 30% of the final course grade and will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines. The project will be graded in week 14 with the following grading breakdown:

Section 1 of the project will be the finished product and will be worth 1/3 of the project grade (10% of your final grade). The finished product will be graded on functionality, layout, fabrication techniques and professionalism as outlined in the rubric. Since the students have successfully completed a skills (fabrication and soldering) course, a portion of the fabrication grade will involve soldering all the components on a perforated prototype board.

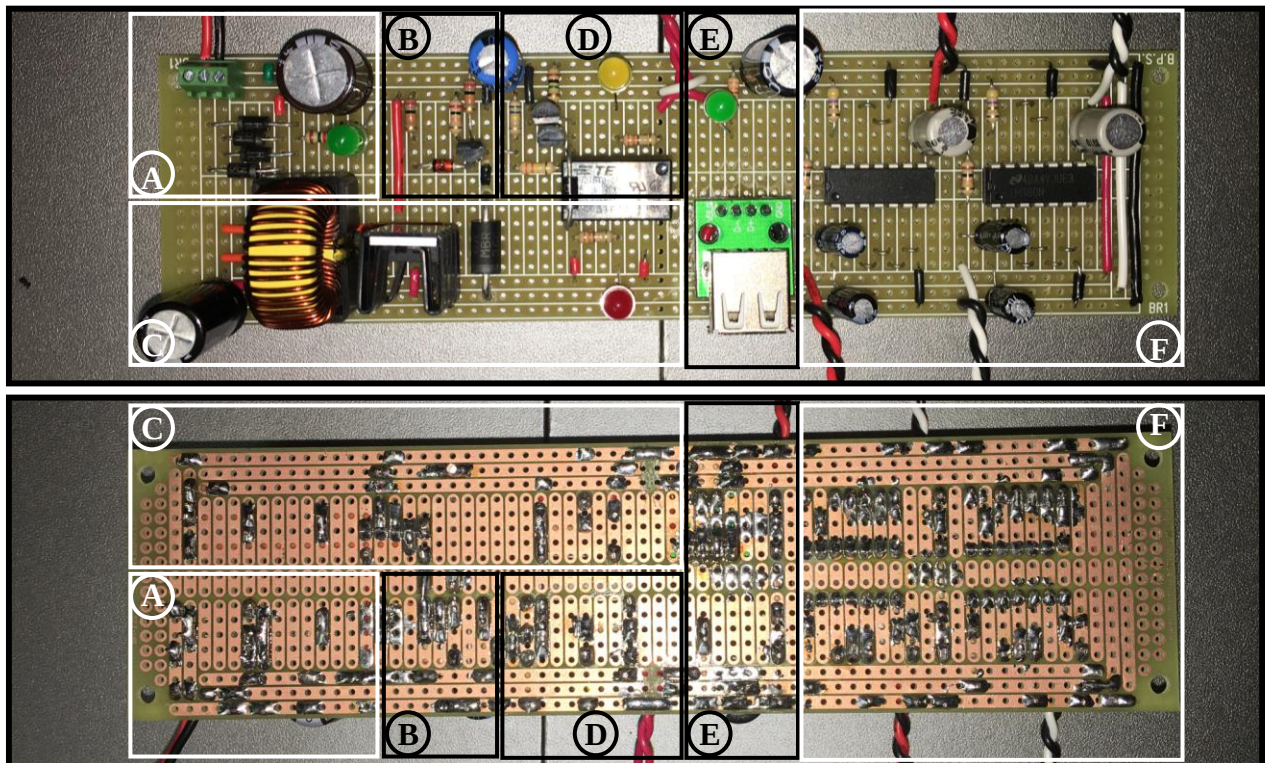
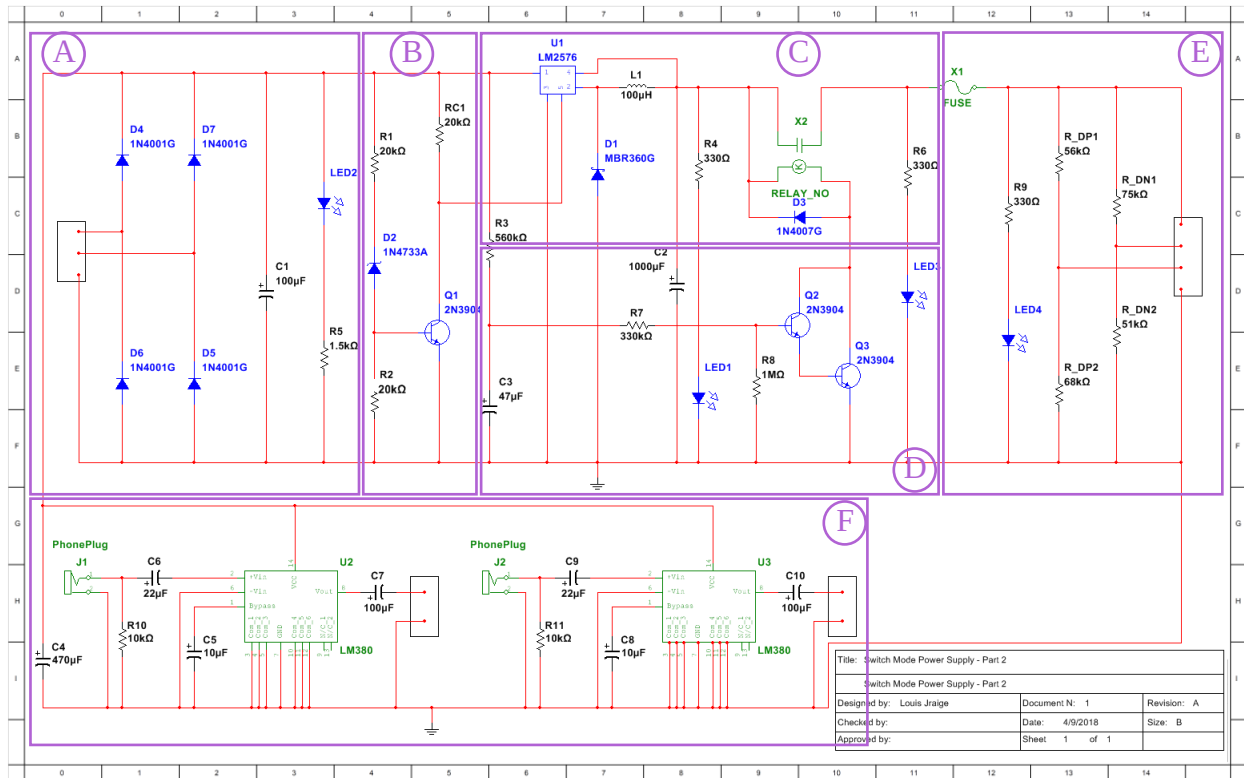
Section 2 of the project will be an evaluation of the demonstration of the performance characteristic measurements of the project and will be worth 1/3 of the project grade (10% of your final grade). For example, some of the performance characteristic measurements for the power supply portion might include line and load regulation, ripple and ripple factor, etc. The demonstration of the performance characteristic measurements will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines.

The third and final portion (Section 3) of the project will be in the written lab report and is worth the remaining 1/3 of the grade (10% of your final grade). The lab report will be written with the guidelines as described below and graded against the rubric below. The written lab report portion will be graded as outlined in the rubric and in accordance with the CVLOs and EESs as outlined in our program mapping and listed in the ministry guidelines.

Section 1 – Layout, Build/Fabrication and Testing

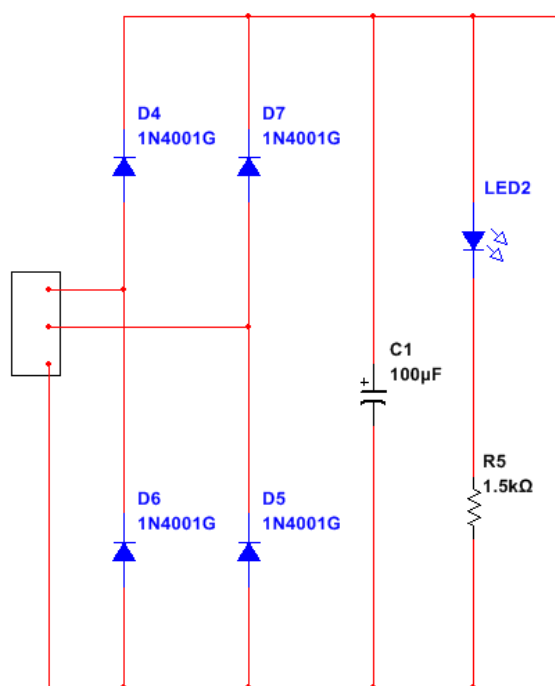
The Schematic Diagram Explained

The following is a phase-by-phase breakdown of the operation of the power supply and charging portion of the project.

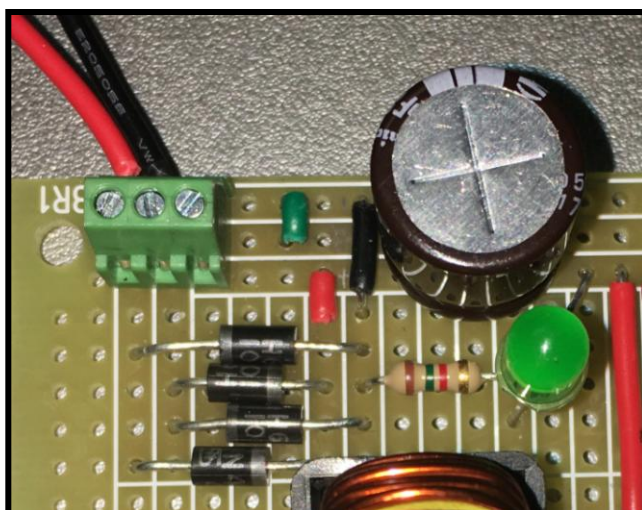


The following is a phase-by-phase breakdown of the operation of the power supply and charging portion of the project. Be sure to carefully layout your circuit before soldering so as not to run out of space on your board. Do **NOT** copy my layout or design. The following is your for your reference only.

Section A – Bridge Rectifier Circuit



This section rectifies and filters the 12 Vac sinusoidal input voltage and then turns on a green LED indicator to signify that there is input power available and ready for use.



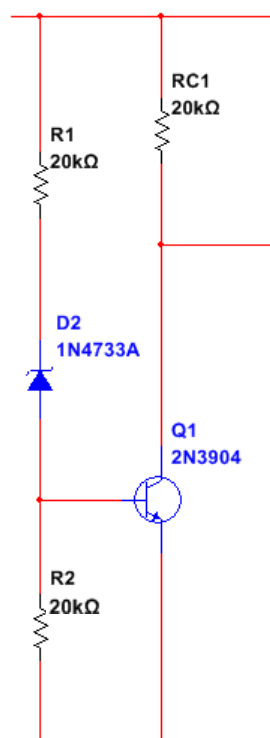
<u>Section</u>	<u>Description</u>	<u>Requirement</u>	<u>Weight</u>
Section A	Bridge Rectifier Circuit	Mandatory	1.5% of 10%

Time: Design: _____ Build: _____ Debug: _____ Data: _____ Document: _____

Bypass Instructions: *There are no bypass instructions as this section is mandatory.*

Lessons Learned: _____

Section B – Under-Voltage Detector Circuit



This section is an “active low” under-voltage monitoring and cut-off circuit. The transistor monitors the regulators input voltage and brings the “Enable” or “On/Off” input to a “logic high” if the supply voltage drops below the voltage across the Zener diode “D2”. Otherwise, the voltage across the resistor “R2” is high enough to drive the transistor in to “saturation” and bring the “Enable” or “On/Off” input to a “logic low”.



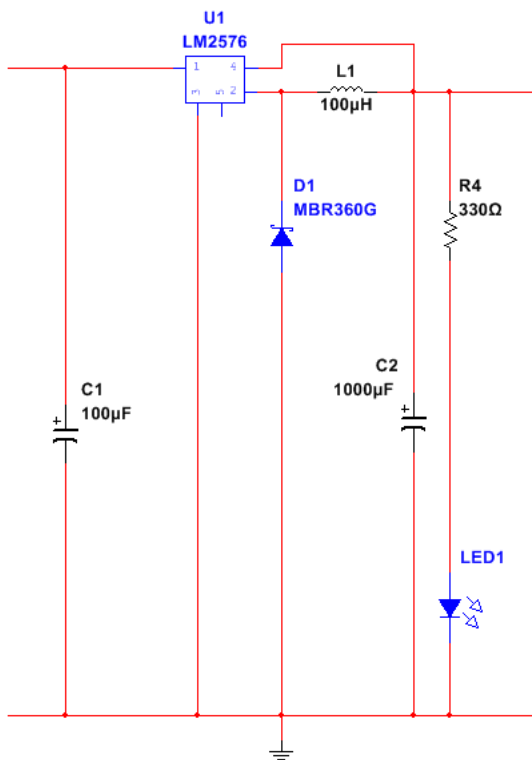
<u>Section</u>	<u>Description</u>	<u>Requirement</u>	<u>Weight</u>
Section B	Under-Voltage Detector Circuit	Optional	1.5% of 10%

Time: Design: _____ Build: _____ Debug: _____ Data: _____ Document: _____

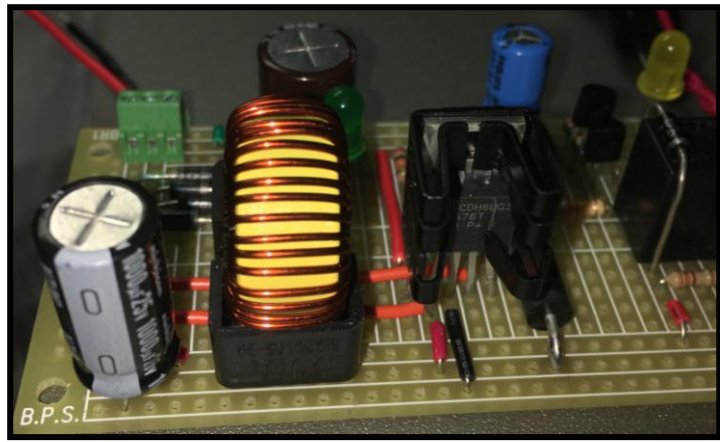
Bypass Instructions: *To bypass this circuit, simply remove R_1 , D_2 , R_2 , R_{C1} and Q_1 . Short pin 5 of the voltage regulator IC (in the next section) to ground to keep it permanently enabled.*

Lessons Learned: _____

Section C – 5 VDC Regulation Circuit



This section is the voltage regulator section. Here the rectified and filtered input voltage (across C1) is brought to 5V_{DC} (by the voltage regulator IC) and made available to the charging circuit across capacitor C2. The red LED indicator signifies that the 5 VDC regulator circuit is functioning properly and that power is available and ready for use.



<u>Section</u>	<u>Description</u>	<u>Requirement</u>	<u>Weight</u>
Section C	Voltage Regulator Circuit	Mandatory	1.5% of 10%

Time: Design: _____ Build: _____ Debug: _____ Data: _____ Document: _____

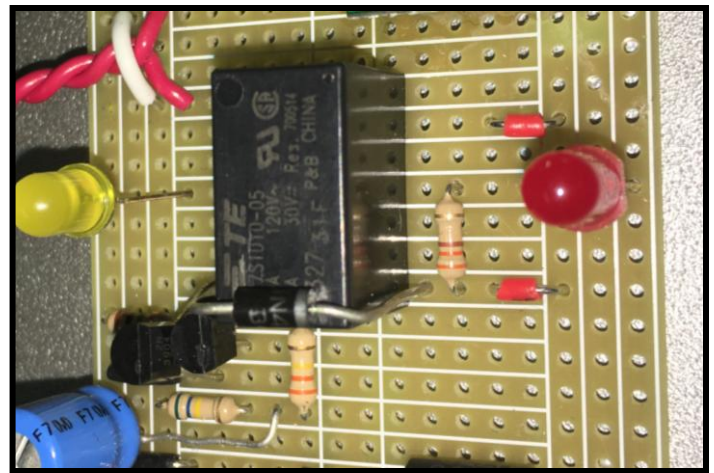
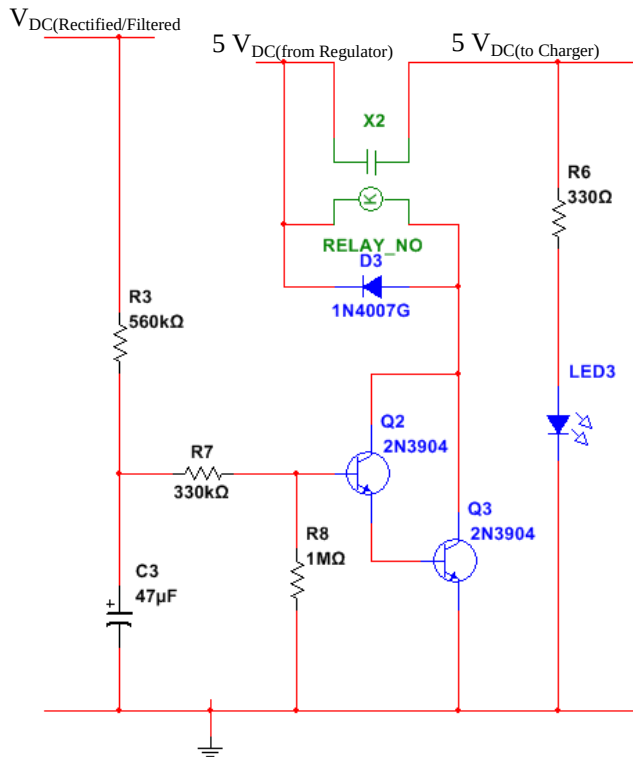
Bypass Instructions: *There are no bypass instructions as this section is mandatory.*

Lessons Learned: _____

Section D – Start Up Delay Circuit

This section is a “start up delay” circuit. The Darlington pair configuration of transistors monitors the voltage divider across the charge timer capacitor and then energizes the relay (by connecting it to ground) when the capacitor builds a significant enough charge across it. The

“fly-back” diode across the relay protects the transistors from a large negative spike when its state changes from open to close. The amber LED indicator signifies that the relay circuit is functioning properly and that power is available and ready for use.



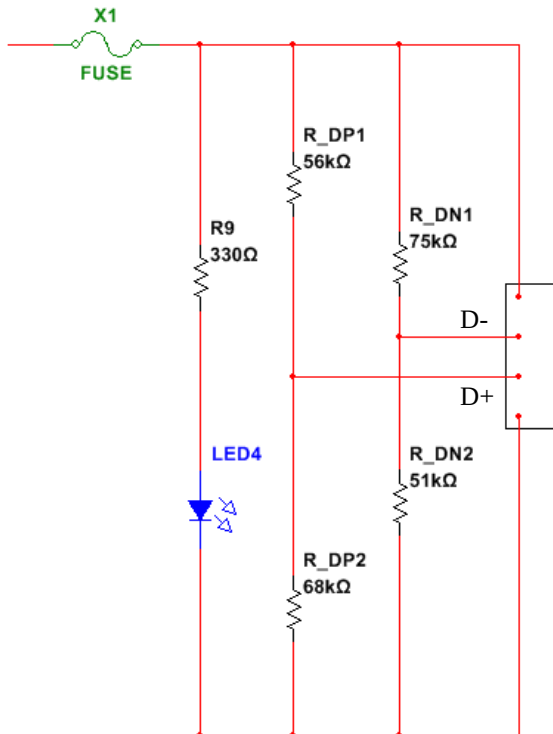
Section	Description	Requirement	Weight
Section D	Start Up Delay Circuit	Optional	1.5% of 10%

Time: Design: _____ Build: _____ Debug: _____ Data: _____ Document: _____

Bypass Instructions: To bypass this circuit, simply remove R_3 , C_3 , R_7 , R_8 , X_2 , Q_2 , Q_3 , R_6 and LED_3 . Simply connect the fuse (in the next section) between the 5 VDC output of the voltage regulator and the indicator (between R_6 and R_9).

Lessons Learned: _____

Section E – 5 VDC USB Charging Circuit



This section is the charging circuit. The incoming power, through the fuse, is supplied to the USB connector. The green LED indicator signifies that the fuse is not damaged. The voltage dividers for “Data -” and “Data +” can be set for the following charging currents.

For a charging current of up to 500 mA @ 2.5W

$R_{DN1} = 75\text{ k}\Omega$, $R_{DN2} = 51\text{ k}\Omega$, $R_{DP1} = 75\text{ k}\Omega$, $R_{DP2} = 51\text{ k}\Omega$

***For a charging current of up to 1 A @ 5W**

$R_{DN1} = 75\text{ k}\Omega$, $R_{DN2} = 51\text{ k}\Omega$, $R_{DP1} = 56\text{ k}\Omega$, $R_{DP2} = 68\text{ k}\Omega$

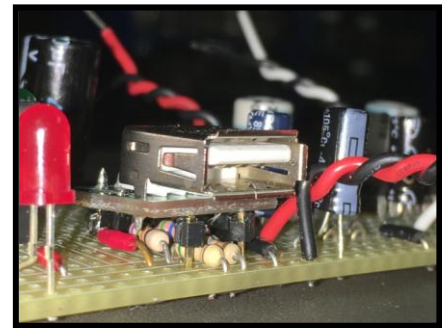
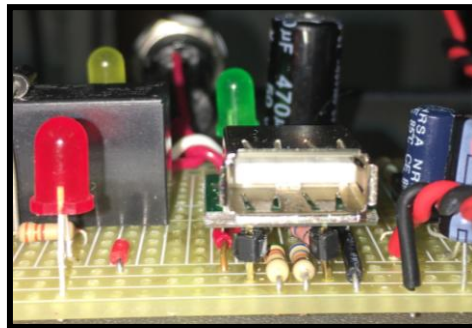
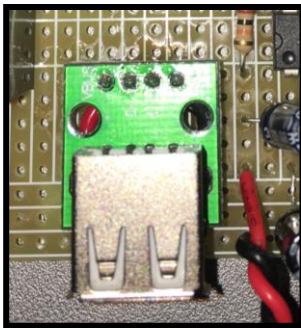
For a charging current of up to 2.1A @ 10W

$R_{DN1} = 56\text{ k}\Omega$, $R_{DN2} = 68\text{ k}\Omega$, $R_{DP1} = 75\text{ k}\Omega$, $R_{DP2} = 51\text{ k}\Omega$

For a charging current of up to 2.4 @ 12W

$R_{DN1} = 56\text{ k}\Omega$, $R_{DN2} = 68\text{ k}\Omega$, $R_{DP1} = 56\text{ k}\Omega$, $R_{DP2} = 68\text{ k}\Omega$

The resistors are not visible in the photo to the right as they are resting on the board under the USB jack. The USB board is suspended on some “stand-offs” so as not to short out on anything.



Section	Description	Requirement	Weight
Section E	5 VDC USB Charging Circuit	Mandatory	1.5% of 10%

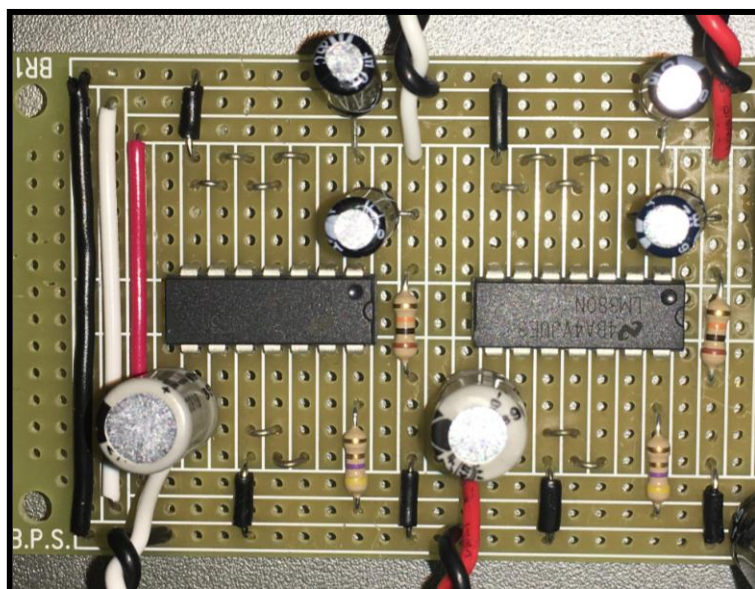
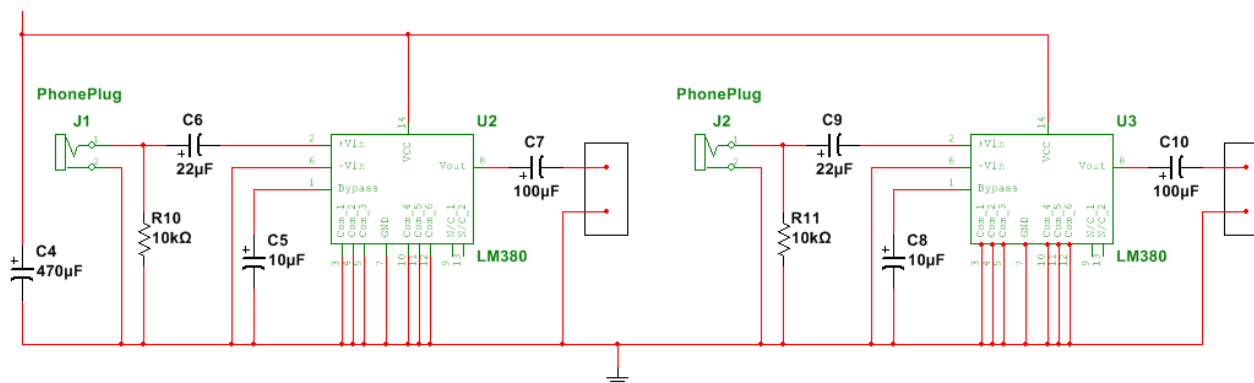
Time: Design: _____ Build: _____ Debug: _____ Data: _____ Document: _____

Bypass Instructions: *There are no bypass instructions as this section is mandatory.*

Lessons Learned: _____

Section F – Audio Amplifier Circuit

This is the audio amplification section. There are two identical circuits; one for each channel (left and right) however only one is shown below. The audio signal is dropped across the input resistor and fed to the amplifier IC through the input coupling capacitor. The amplified output signal is coupled to the speaker through the output coupling capacitor.



Although both a right and left channel are shown above and to the left, only one of them is mandatory. This means that you can choose not to build, test, measure and document one side only.

Please ignore the resistor shown connected to pin 14 of the IC. Replace it with the jumper shown in the schematic diagram. They were added in the design phase and never removed in the simplified circuit design.

Section	Description	Requirement	Weight
Section F	Audio Amplifier Circuits	Mandatory/Optional	1.0% of 10%

Time: Design: _____ Build: _____ Debug: _____ Data: _____ Document: _____

Bypass Instructions: Simply build only one of the output channels.

Lessons Learned: _____